Machine Learning

06-04-26

Chain rule of derivatives

if $y = f(z)$ $\quad z = f(p)$ $\quad p = f(x)$

then

$$\frac{\partial y}{\partial x} = \frac{\partial y}{\partial z} \times \frac{\partial z}{\partial p} \times \frac{\partial p}{\partial x}$$

$$y = \{8 - [(x+2)^2 + 5]^3\}^5$$

$z = 8 - [(x+2)^2 + 5]^3$ $\quad p = (x+2)^2 + 5$ $\quad q = x+2$

$y = z^5$ $\quad z = 8 - p^3$ $\quad p = q^2 + 5$

$$\frac{\partial y}{\partial x} = \frac{\partial y}{\partial z} \cdot \frac{\partial z}{\partial p} \cdot \frac{\partial p}{\partial q} \cdot \frac{\partial q}{\partial x}$$

$$= 5z^4 \times (0 - 3p^2) \times (2q + 0) \times (1 + 0)$$

$$\frac{\partial y}{\partial x} = 5z^4 \times -3p^2 \times 2q \times 1$$

$$\boxed{\frac{\partial y}{\partial x} = -30 z^4 p^2 q}$$

Gradient Descent

- its method of finding minima of function using its slope/ derivatives.

hypothesis function : $y = wx + b$

dataset

x	y
1	3
2	4

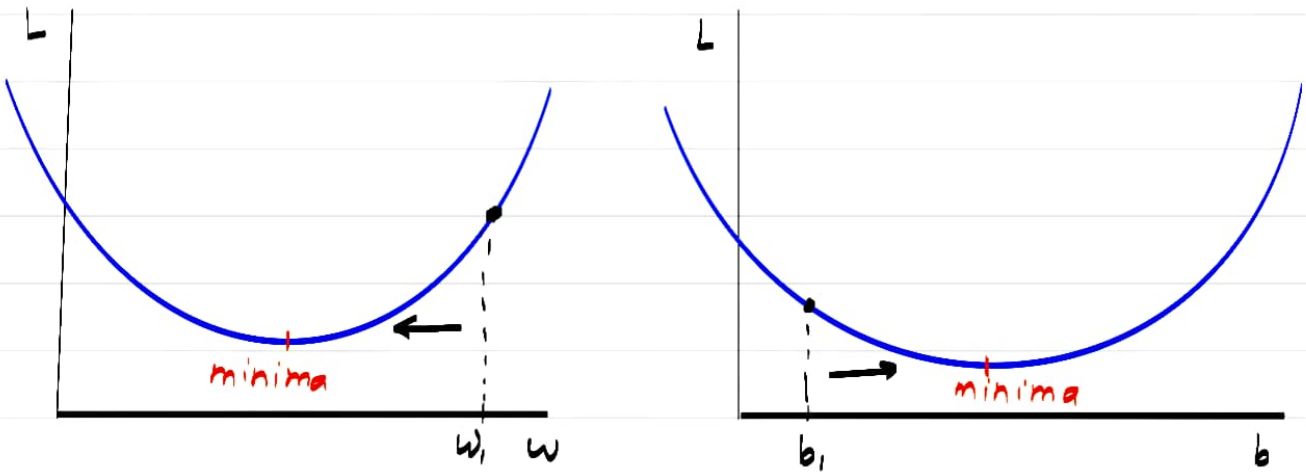
Loss function : $L = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$

finding minima $\longrightarrow$ finding optimal values of w and b giving min error/loss

we start with random initialization

$w = w_1$ and $b = b_1$

eg. $w = 1$ $b = 0$ / $w = 2$ $b = 5$



as w_1 and b_1 are not minima parameters, we need to update them to find minima.

change value of w and b to move towards minima

- reduce value of w
- increase value of b

How to update w and b?

We'll update w and b using their gradients = slopes.

$$L = f(w)$$

$$\frac{\partial L}{\partial w}$$

x	y
1	3
2	4

$$L = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$n = 2$$

$$L = \frac{1}{2} \left\{ [y_1 - \hat{y}_1]^2 + [y_2 - \hat{y}_2]^2 \right\}$$

$$= \frac{1}{2} \left\{ [3 - (wx_1 + b)]^2 + [4 - (wx_2 + b)]^2 \right\}$$

$$= \frac{1}{2} \left\{ [3 - (w + b)]^2 + [4 - (2w + b)]^2 \right\}$$

$$= \frac{1}{2} \left\{ 9 + (w+b)^2 - 6(w+b) + 16 + (2w+b)^2 - 8(2w+b) \right\}$$

$$= \frac{1}{2} \left\{ 25 + w^2 + b^2 + 2wb - 6w - 6b + 4w^2 + b^2 + 4wb - 16w - 8b \right\}$$

$$= \frac{1}{2} \left\{ 5w^2 + 2b^2 + 6wb - 22w - 14b + 25 \right\}$$

$$L = \frac{5}{2} w^2 + b^2 + 3wb - 11w - 7b + \frac{25}{2}$$

$$\frac{\partial L}{\partial w} = \frac{5}{2} \times 2w + 0 + 3b - 11 - 0 + 0$$

$$\frac{\partial L}{\partial w} = 5w + 3b - 11$$

current values of $w = w_1 = 1$ and $b = b_1 = 0$

update w using $\frac{\partial L}{\partial w}$

$$w_2 = w_1 - \alpha \frac{\partial L}{\partial w}$$

α - learning rate

$$w_2 = w_1 - \alpha \frac{\partial L}{\partial w_1}$$

$$w_2 = w_1 - \alpha(-6)$$

$$w_2 = 1 - \alpha(-6)$$

α → learning rate

a small +ve constant

$\alpha = 0.01,\ 0.001,\ 0.1,\ 10^{-4}$

$$\frac{\partial L}{\partial w_1} = 5w_1 + 3b_1 - 11$$

$w_1 = 1 \quad b_1 = 0$

$$\frac{\partial L}{\partial w_1} = 5\times 1 + 3\times 0 - 11 = -6$$

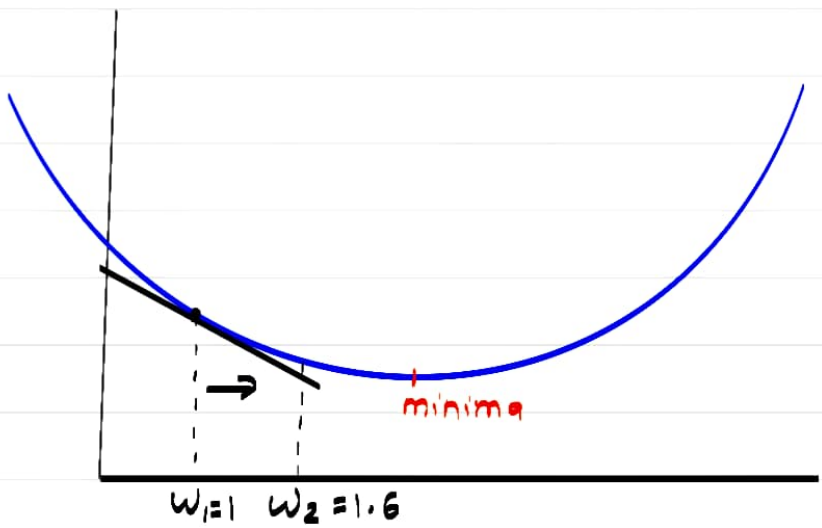
$$\frac{\partial L}{\partial w_1} = -6$$

$$\frac{\partial L}{\partial w_1} < 0$$

lets put $\alpha = 0.1$

$$w_2 = 1 - 0.1(-6)$$

$$w_2 = 1.6$$



Now let's find $\frac{\partial L}{\partial b}$ for updating value of b

$$L = \frac{5}{2}w^2 + b^2 + 3wb - 11w - 7b + \frac{25}{2}$$

$$\frac{\partial L}{\partial b} = 0 + 2b + 3w - 0 - 7 + 0$$

$$\frac{\partial L}{\partial b} = 2b + 3w - 7$$

$$\frac{\partial L}{\partial b_1} = 2\times b_1 + 3\times w_1 - 7 = 0 + 3 - 7 = -4$$

$$b_2 = b_1 - \alpha \frac{\partial L}{\partial b_1}$$

$$= 0 - 0.1 \times (-4)$$

$$b_2 = 0.4$$

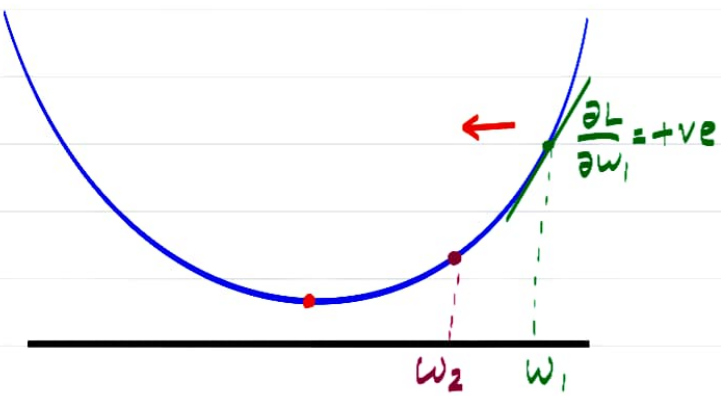
L at w_2 < L at w_1

L at b_2 < L at b_1

Parameter updation rule

$$w_{new} = w_{old} - \alpha \frac{\partial L}{\partial w_{old}}$$

$$b_{new} = b_{old} - \alpha \frac{\partial L}{\partial b_{old}}$$



$$w_2 = w_1 - \alpha \frac{\partial L}{\partial w_1}$$

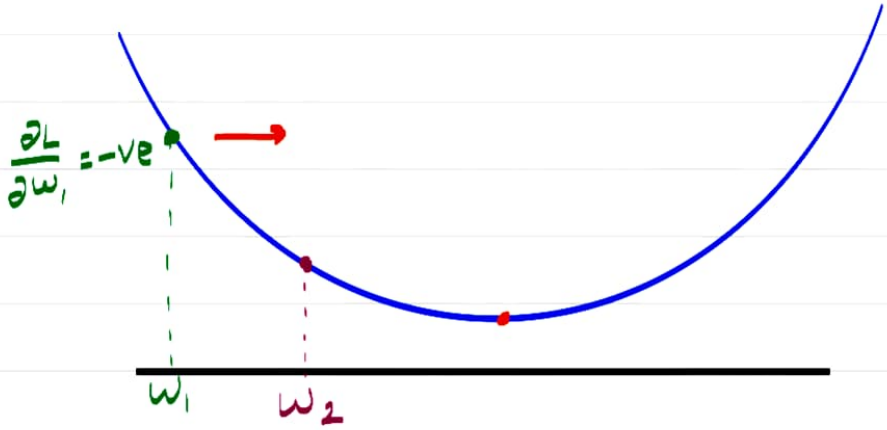
lets say $w_1 = 2$ $\frac{\partial L}{\partial w_1} = 1.5$ $\alpha = 0.1$

$$w_2 = 2 - 0.1 \times (1.5)$$

$$= 2 - 0.15 = 1.85$$

$$w_2 = 1.85$$

$\Rightarrow$ by applying parameter updation rule we've moved towards minima



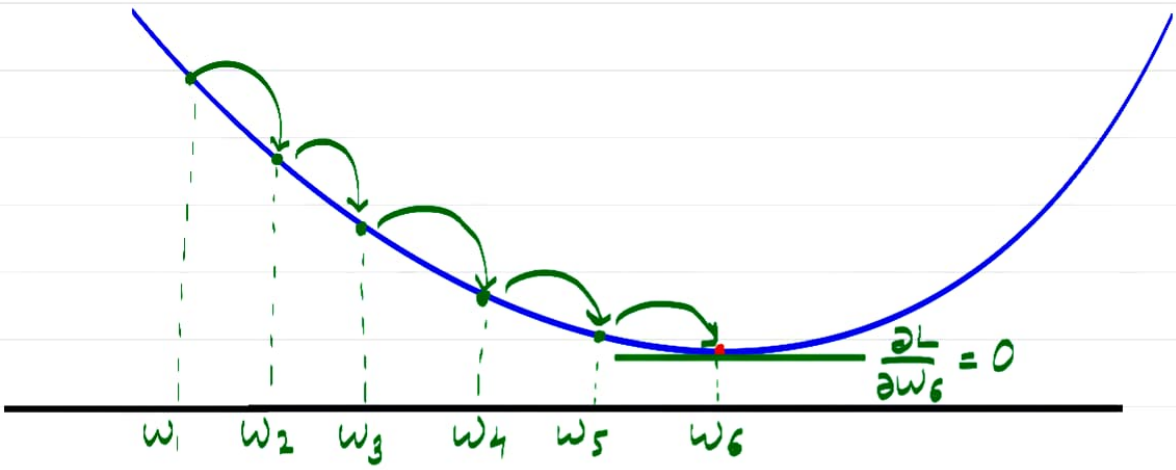
$$w_2 = w_1 - \alpha \frac{\partial L}{\partial w_1}$$

lets say $\quad w_1 = 1 \quad \frac{\partial L}{\partial w_1} = -0.5 \quad \alpha = 0.1$

$$w_2 = 1 - 0.1 \times (-0.5)$$
$$= 1 + 0.05$$
$$w_2 = 1.05$$

here also we've moved towards minima

- parameter updation using gradient descent always insures movement towards minima.



$w_2 = w_1 - \alpha \frac{\partial L}{\partial w_1}$ $\quad$ $w_4 = w_3 - \alpha \frac{\partial L}{\partial w_3}$ $\quad$ $w_6 = w_5 - \alpha \, \partial L / \partial w_5$

$w_3 = w_2 - \alpha \frac{\partial L}{\partial w_2}$ $\quad$ $w_5 = w_4 - \alpha \frac{\partial L}{\partial w_4}$ $\quad$ $w_7 = w_6 - \alpha \, \partial L / \partial w_6$

as $\partial L / \partial w_6 = 0$

$w_7 = w_6$

- eventually we'll reach at minima where parameter updation will stop.

- during model training, we keep updating model parameters w and b until we reach minima.

How do we get to know that we have reached minima?

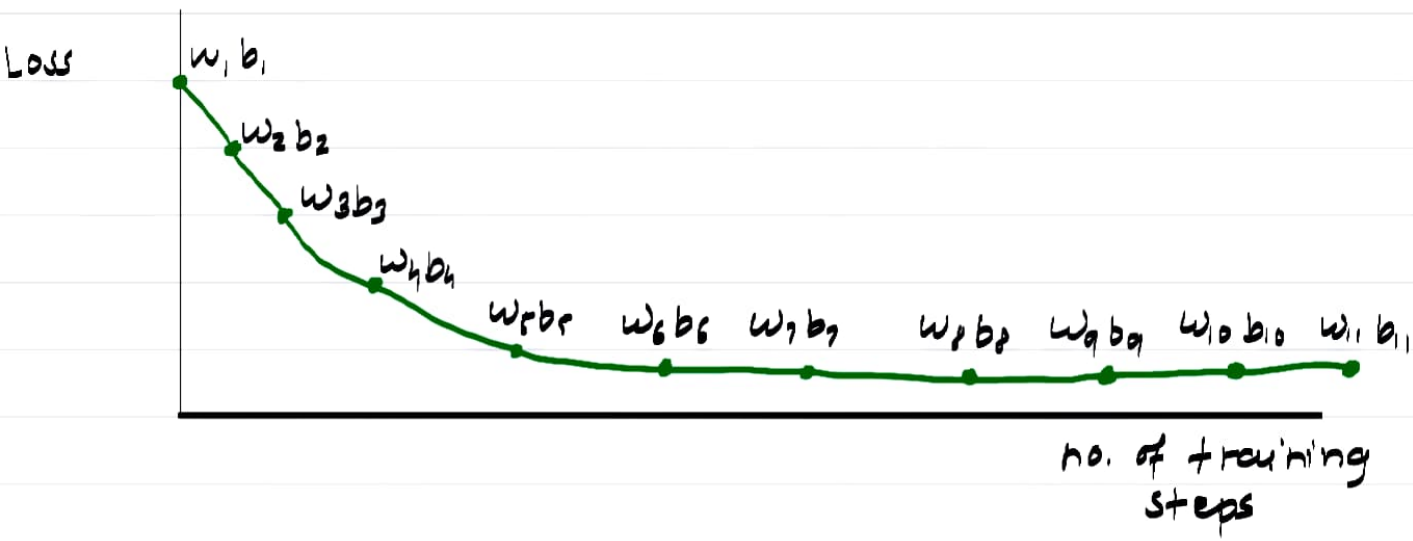
- the slope becomes zero

$$\frac{\partial L}{\partial w} \approx 0 \qquad \frac{\partial L}{\partial b} \approx 0$$

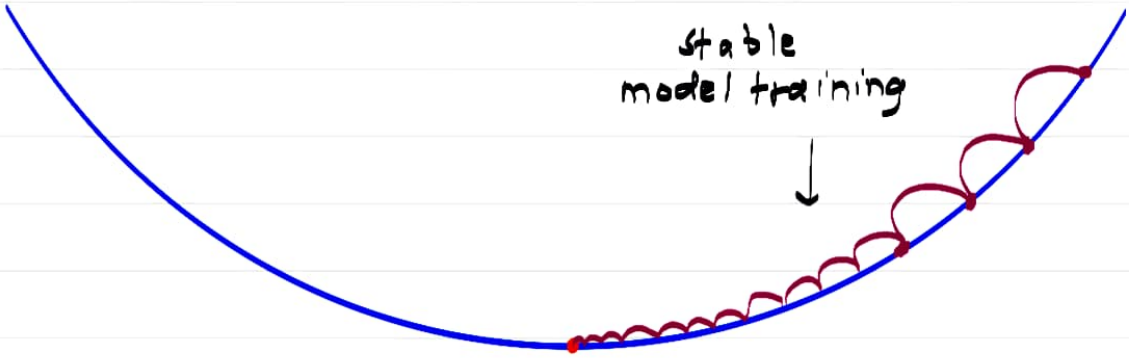
- parameter updation stops

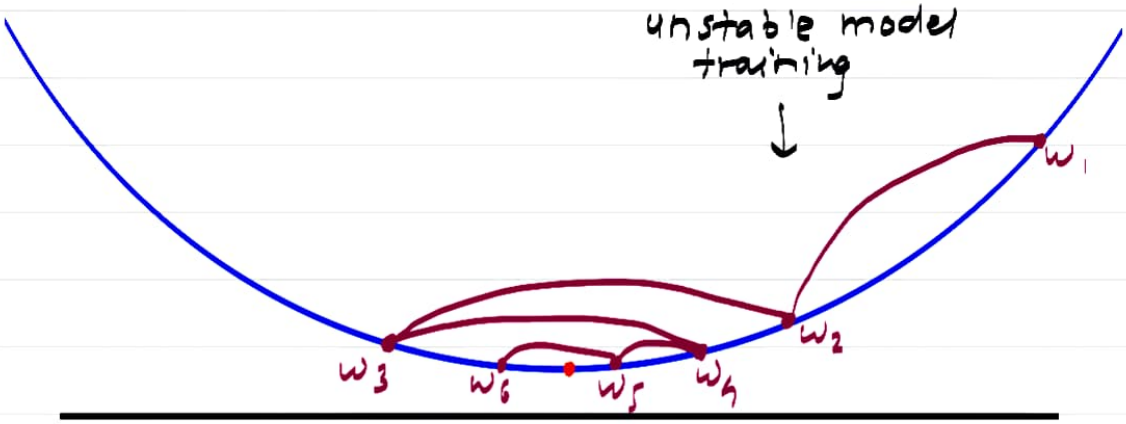
$$w_{new} \approx w_{old} \qquad b_{new} \approx b_{old}$$

- As model parameters updation stops, loss value doesn't decrease further



Learning rate (α)





- At start gradient values are large and model takes bigger steps towards minima

- In some cases, if slope value is very large, then it might overshoot minima

- to counter large values of slope, we use learning rate

- learning rate makes model training stable

LR $= \alpha$ is a hyperparameter

hence its best and optimal value is found using hyperparameter tuning